

Agent Profile Template

One sheet per agent, and the three checks a pair of profiles has to pass

1 Whose words to use

Question 44 and question 38 have one answer between them. The security community already has a settled vocabulary for both halves, attacker and defender, and it is not the vocabulary the reinforcement learning side uses. A reviewer who works in security reads our words as looseness, and they are right to. This table is the mapping.

What we say	What they say	Where it is fixed
Opponent type	Threat actor, characterized by sophistication and resource level as two separate properties	STIX 2.1, Threat Actor object
Scripted opponent	<i>Adversary emulation</i> when it reproduces a documented actor, <i>red teaming</i> when it searches freely for a way through	MITRE ATT&CK, and this project's own threat modeling section
Opponent action set	Tactics, techniques, and procedures	ATT&CK
Defender priorities	Security posture, set by a risk appetite and made operational by a risk tolerance	CNSSI 4009-2022, and NIST CSF 2.0 at GV.RM-02
Defender action set	Defensive tactics: Model, Harden, Detect, Isolate, Deceive, Evict, Restore	MITRE D3FEND
Deception actions	Adversary engagement, with goals Expose, Affect, and Elicit	MITRE Engage

One term to stop using loosely. Question 38 says attacker emulation. The term of art is *adversary emulation*, and it means something narrower than we mean. An emulation plan starts from cyber threat intelligence on a named actor and reproduces not only what that actor does but which tools and commands it does it with and at which stage of a breach. Emulation is a claim about provenance, not a claim about fidelity. A profile invented for a testbed can be sharper, more separable and better controlled than any documented actor, and it is still not emulation.

2 The attacker sheet

Nine fields. Every one of them is short. A field left blank is a field a reviewer fills in for you.

Name and class. The label used in every table and figure, fixed once.

Sophistication. An ordered scale, running from a complete novice to an actor able to influence supply chains in order to introduce vulnerabilities. STIX runs it as none, minimal, intermediate, advanced, expert, innovator, strategic.

Resource level. A second axis, deliberately separate from the first: individual, club, contest, team, organization, government. They are separate because a highly skilled actor may have almost nothing behind it while a minimal one may have the resources of an organized crime ring.

Goals. What this actor wants out of this network, named in the network's own objects rather than in reward.

Primary motivation. One value. Secondary motivations go on a second line or nowhere.

Techniques. The action set, mapped to ATT&CK wherever a mapping exists, and marked as unmapped where none does.

Decision rule. What the actor does with what it sees. This is the field that makes it an agent instead of a trace, and it is the field the opponent model is trying to recover.

Observability. What the actor can see. Probes belong here, with what each one returns.

Provenance. A documented actor and the intelligence that documents it, or invented for this testbed and the behavior it abstracts. One or the other. There is no third answer, and writing neither is how a paper ends up claiming emulation it did not do.

3 The defender sheet

Question 44 wants defenders with different priorities and different postures. Posture is the right word and it has a definition: “the security status of an enterprise’s networks, information, and systems based on CS resources (e.g., people, hardware, software, policies) and capabilities in place to manage the defense of the enterprise and to react as the situation changes.” Six fields.

Posture, in one sentence. What this defender is protecting and what it is willing to lose to protect it.

Risk appetite, then risk tolerance. Appetite is how much risk the organization will accept in pursuit of an objective. Tolerance is the operational limit inside that appetite, and in an experiment it is a number in the reward function. Write the number first and the reward second. Doing it the other way around means the posture is whatever the reward happened to imply.

Permitted tactics. Which of the seven defensive tactics this profile may use at all. Two agents that differ only in the weights on one reward are the same profile trained twice. Two agents that differ in which tactics they may reach for are two profiles.

Engagement goal. Evict, or elicit. On one host, at one time, it cannot be both, and choosing between them is the sharpest difference there is between two real defenders. It is also the choice the deception action makes.

Observability. What it sees and at what rate. Verbosity is an action in most action sets, so state the default and state what raising it costs.

What it will not do. The action this profile declines, and the reason. A profile with nothing in this field is an optimizer, not a posture.

The action space, said in their words

Our action	D3FEND tactic	What it commits the defender to
Isolate	Isolate	A barrier, at the cost of the traffic behind it
Disrupt	Evict, Restore	Removal of in-memory persistence, and of live sessions
Monitor	Detect	Fidelity later, paid for now
Deceive	Deceive	Engagement, which is Expose if the alert is the point and Elicit if the attacker’s next technique is
Null	None	The decision to spend nothing this step

The two tactics with no action in this space are Model and Harden, and both of them happen before an episode starts. Saying so in the threat model costs a sentence. Being asked for it costs a review cycle.

4 The three checks

These are what question 38 asks for when it says double check our justification. Run them on the pair, not on either profile alone.

1. Name the dimension before you name the profiles. Two profiles are a pair only if you can say in one clause what varies between them. If the clause is hard to write, the pair was chosen by what was easy to script.

2. The separating dimension must not be the one the method reads. This is what Reviewer C asked, in C8: “The attacker profiles may favor the proposed opponent model. Avoidant and Enticed mainly differ in how they respond to deception signals, which directly matches the defender’s deception actions. Could the authors evaluate attackers that differ in other behaviors to show that the benefit of opponent modeling is not specific to this setup?” When the dimension that separates the profiles is the dimension the encoder is built to recover, the experiment measures the pair rather than the method. The fix is a second pair varying along something the encoder does not see, reported in the same table as the first.

3. A bracketing claim needs the study that fixes the bracket. Saying two profiles bracket a spectrum observed in field studies is a claim about the field studies, and it is checkable. Either cite them and name both endpoints in their terms, or drop the word and say what is true instead: the profiles span the decision rule applied to deception outcomes, and nothing else.

5 The macros

Paste these into the preamble of whatever the profiles are being written into. They mark the fields the coverage check looks for, and they typeset as plain text.

```
% One sheet. #1 attacker or defender, #2 the profile name.  
\newcommand{\apsheet}[2]{\subsection*{#1 profile: #2}}
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% The fields. Kept as macros so a missing one is greppable.
\newcommand{\apfield}[2]{\textbf{\#1.} \#2}

% The separating dimension of a pair, stated once, before either profile.
\newcommand{\apdim}[1]{\#1}

% Provenance. Documented actor plus source, or invented plus what it abstracts.
\newcommand{\apsrc}[1]{\#1}

% The action the profile declines, and why. Blank fails the defender sheet.
\newcommand{\apdecline}[2]{\#1, because \#2}

% Risk tolerance as a number, written before the reward that encodes it.
\newcommand{\aptol}[2]{\#1 per \#2}

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6 Filled, on this project

The attacker pair. Avoidant and Enticed sit at the same point on any sophistication scale, because both run the same three deterministic binary probes over the same attacker-observable information. Resource level is not stated for either and does not need to be, since neither profile spends anything. The separating dimension is the decision rule applied to the probe outcomes, conservative against permissive. Provenance is invented for this testbed, abstracting the honeypot fingerprinting work the banner probe is built on, and it is not emulation of a named actor. Check 2 is the one that costs us: the separating dimension is exactly the signal the opponent encoder reads, which is why C8 landed.

The defender pair that would answer it. An evicting defender isolates and disrupts on the first alert that clears threshold, never deceives, and accepts dropped sessions as the price. An eliciting defender leaves the honeypot standing while the attacker is inside it, pays the alert cost, and declines to isolate until the attacker has revealed a second technique. They differ in engagement goal, in permitted tactic, and in tolerance for user-visible disruption, and that third one has to be a number in the reward before either is trained. Neither of them differs from the other in a way the opponent encoder was designed to read, which is the point.

7 Sources

- The threat actor property set, the sophistication scale and its endpoints, the resource level scale, and the argument for keeping the two apart are from the STIX 2.1 specification, Threat Actor object and its vocabularies.
- The adversary emulation definition, the emulation plan structure (intelligence summary, operations flow, step by step procedures in human and machine readable form, infrastructure), and the split between full and micro plans are from MITRE ATT&CK and the Center for Threat-Informed Defense adversary emulation library.
- The seven defensive tactics are MITRE D3FEND. The engagement goals Expose, Affect, and

Elicit, and the surrounding Prepare and Understand, are MITRE Engage.

- The security posture definition quoted in section 3 is CNSSI 4009-2022, as carried in the NIST glossary. The risk appetite and risk tolerance distinction is NIST CSF 2.0 GV.RM-02 and NIST IR 8286A.
- The separation between adversary emulation and red teaming used here is the stipulative one already written into this project's own threat modeling section, where emulation is the narrower case that models a particular adversary's known behavior.
- NIST SP 800-30 Rev. 1 assesses an adversarial threat source along capability, intent, and targeting. Read through a search summary rather than from the publication, so it is marked secondary and should be confirmed against Appendix D before being cited.
- C8 is quoted from the review email of 19 August 2026. The probe description, the profile decision rules, and the five element defender action space are from the threat model and testbed section of the paper.